ZCAS University

CCS4621 ADVANCED NETWORKING

END OF SEMESTER EXAMINATION

18TH JUNE, 2025

09:00 HRS-12:00HRS

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

You are interested in knowing the protocols running on a network server and decide to do a tcp scan. Consider the exhibit of network scan below and answer the questions that follow.

```
Host is up (0.060s latency).
Not shown: 987 filtered ports
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
80/tcp   open  http
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
443/tcp  open  https
445/tcp  open  microsoft-ds
512/tcp  open  exec
514/tcp  open  shell
992/tcp  open  telnets
3306/tcp open  mysql
5357/tcp open  wsdapi

Read data files from: /usr/bin/../share/nmap
Nmap done: 1 IP address (1 host up) scanned in 9.88 seconds
```

Figure 1. Results of a network security scan

(a) State which protocol is used for unsecured and secured remote access services. **[4 Marks]**

(b) State which protocol is used for unsecured and secured website services. **[4 Marks]**

(c) State which protocol you can used in place of ftp to provide secure file transfer. **[4 Marks]**

(d) Explain the term tcp appearing after each port number. **[4 Marks]**

(e) Differentiate in tabular form the TCP 3-way handshake from the WiFi 4-way handshake. **[4 Marks]**

Question 2

Study carefully the undirected graph below reflecting a network structure with nodes representative of network hosts and devices, and edges representing interlinks amongst them.

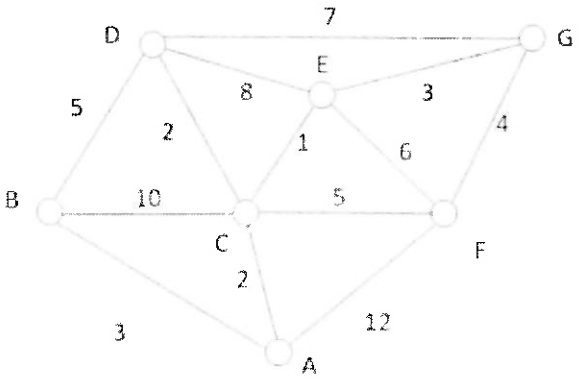


Figure 2. Undirected network structure

(a) State the conditions for a Euler path and Euler circuit and ascertain whether they exist in this network structure. **[5 Marks]**

(b) Compute the incidence matrix **[5 Marks]**

(c) Use the Dijkstra algorithm to find the shortest path from B to G by evaluating all possible routes between the nodes. **[5 Marks]**

(d) Explain which and how the routing algorithms are used in BGP and IGRP to find the best route in a given network structure. **[5 Marks]**

(Total: 40 marks)

SECTION B: Attempt any THREE questions. Each questions carries 20 marks.

Question 3

Consider the network of switches below in Figure 1 where STP is to be implemented and answer the questions that follow:

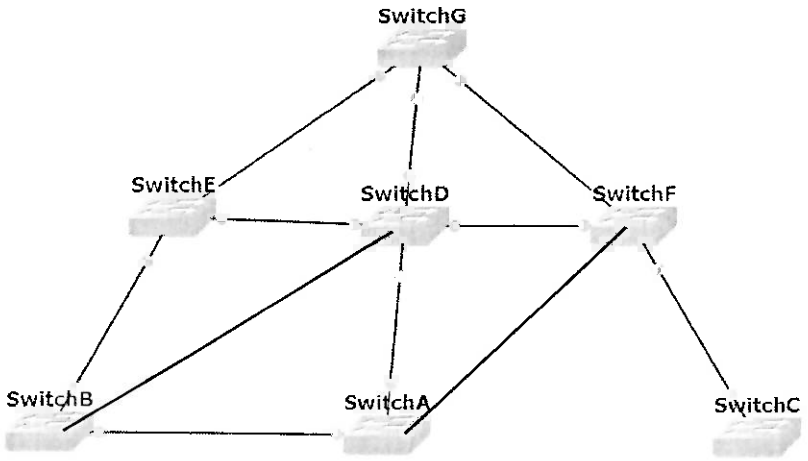


Figure 1. Switched network

i. Determine and state the number of cycles in the network by drawing a graph depicting well-labeled cycles. **[5 Marks]**

ii. Name the switch, which will be chosen as the root bridge and explain why. **[5 Marks]**

iii. Given the switch labels denote their IDs, use the Spanning Tree Protocol to redraw and label all the ports as RP, DP, and BP. **[5 Marks]**

iv. Derive and draw the resultant spanning tree. **[5 Marks]**

Question 4

You are employed as the network administrator of the above network, and you are asked to create subnets using **VLSM** for the IP address range 19.0.0.0/16 for the above network. Subnet accordingly and complete the following table:

	Network ID	**Subnet Mask**	**CIDR Netmask**	**Broadcast ID**	**Wildcard Mask**
Network 1					
Network 2					
Network 3					
Network 4					

[10 Marks]

Question 5

As networks evolve to support growing demands in scalability, security, automation, and flexibility, technologies such as IPv6, Software Defined Networking (SDN), Cloud Computing, the Internet of Things (IoT), and Network Address Translation (NAT/PAT) have emerged as critical enablers of modern digital infrastructure. Understanding these technologies is essential for designing and maintaining robust and future-proof networks.

(a) Explain two key limitations of IPv4 that IPv6 addresses. Then briefly describe how IPv6 improves routing efficiency and scalability in large networks. **[5 Marks]**

(b) Define the architecture of SDN and explain how it differs from traditional networking. Provide one real-world benefit of SDN in managing enterprise or cloud networks. **[5 Marks]**

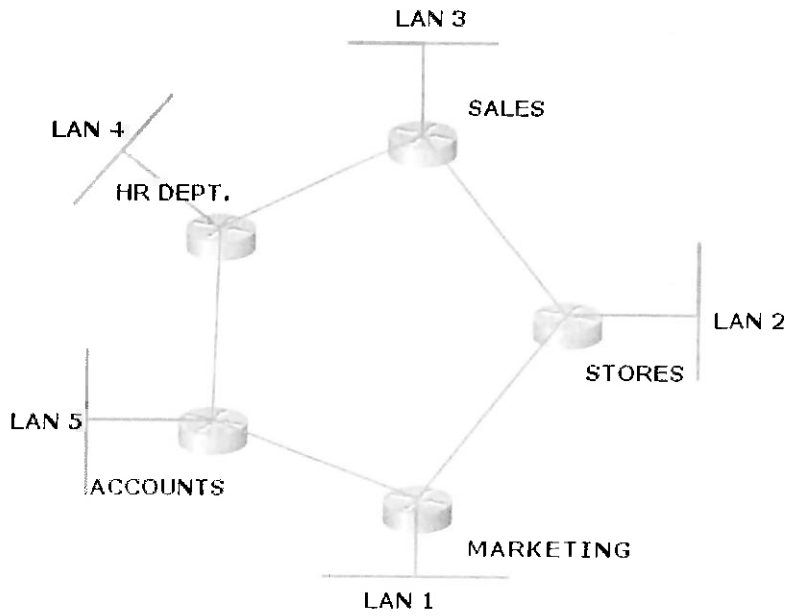
(c) Describe the three primary service models of cloud computing (IaaS, PaaS, SaaS), and explain how each supports different layers of enterprise IT infrastructure. **[5 Marks]**

(d) Differentiate between Network Address Translation (NAT) and Port Address Translation (PAT). Use a simple diagram or example to show how PAT enables multiple devices to share a single public IP. **[5 Marks]**

(20 Marks)

Question 6

1. You are the Network Engineer of a firm that needs to subnet a Class C IP address range of 99.0.0.0/8. You have five departments in your building namely; MARKETING, STORES, SALES, HR DEPT & ACCOUNTS and they are linked together as shown in the network diagram below.



(a) Given that each department has a maximum of 500 hosts, find the total number of collision and broadcast domains in the diagram above. **[5 Marks]**

(b) State the network and broadcast address you will allocate to each department after subnetting the given network address. **[10 Marks]**

(c) State any two command line utilities/commands that you can use to check the network connectivity and state the layer of the OSI model on which they operate. **[5 Marks]**

(20 Marks)

---------- **END OF EXAMINATION** ----------